

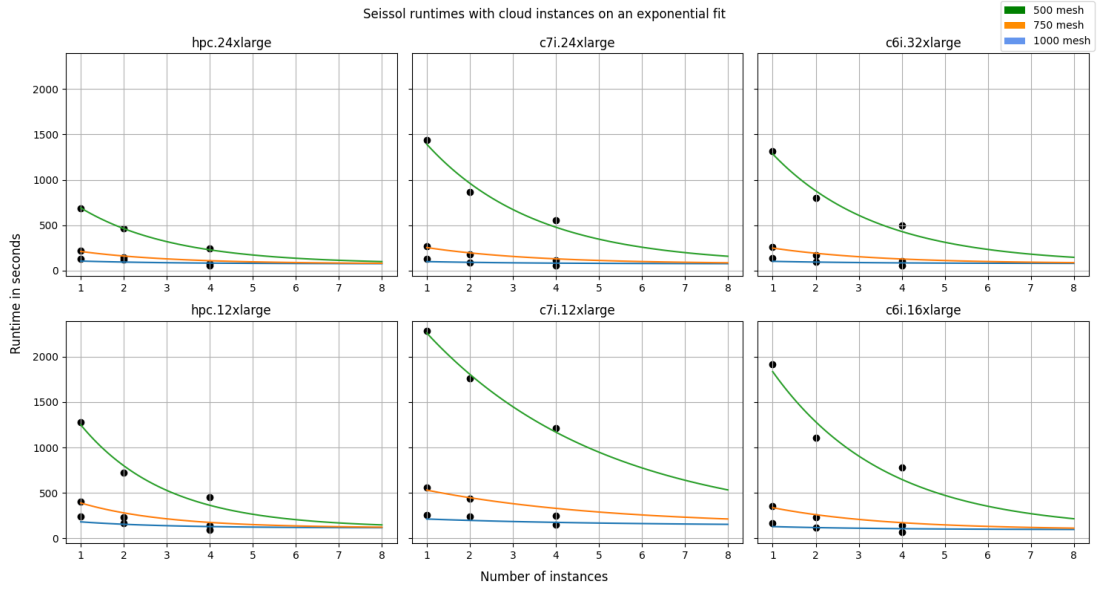


Figure A.3.: Runtime curves for cloud instances

the curve fit presented in Figure 6.2. The equation coefficients for the other instance types and their fit for the exponential runtime is illustrated in Table A.1 and Figure A.3.

Table A.1.: Coefficients (a, b, c, d) and the R^2 score for runtime equations of cloud instances

Instance	a	b	c	d	R^2
hpc7a.24xlarge	1.8e5	-0.50	-0.011	135.4	0.9930
hpc7a.12xlarge	3.2e4	-0.51	-0.006	115.3	0.9801
c7i.24xlarge	1.1e5	-0.39	-0.008	75.8	0.9880
c7i.12xlarge	8.0e4	-0.24	-0.007	141.1	0.9966
c6i.32xlarge	9.1e4	-0.41	-0.008	78.9	0.9895
c6i.16xlarge	1.3e5	-0.38	-0.008	95.8	0.9805

A.3. On-premise Cost Calculation

The cost calculation for a local on-premise cluster was discussed in Section 5.2. Table 5.1 and Table 5.2 tabulated the fixed and variable costs for a cluster. The derivation of the hourly cost of one node of an on-premise cluster from these costs is explained below.

- **Compute Nodes:** Each node in our cluster is estimated at \$12,000 based on comparable pricing of HPC cluster nodes, which include dual Intel Xeon CPUs, memory, and base server infrastructure, scaled from pricing costs from [47].

$$C_{\text{nodes}} = 148 \times 12,000 = \$1,776,000$$

- **Storage:** Based on LRZ's published SuperMUC storage specs [24], the proportional allocation for a 148-node setup includes approximately 240 TB of parallel storage and 56 TB of Network-attached storage.

Cost assumptions:

- Parallel: $240 \text{ TB} \times \$150/\text{TB} = \$36,000$
- NAS: $56 \text{ TB} \times \$100/\text{TB} = \$5,600$

$$C_{\text{storage}} = 36,000 + 5,600 = \$41,600$$

- **Networking:** Full networking costs include switches, network interface cards, and cabling with estimated pricing from [29].

- 5 high-speed switches: $5 \times \$12,000 = \$60,000$
- 148 NICs at \$1,000 each = \$148,000
- DAC optical cabling and installation = \$52,200

$$C_{\text{network}} = 60,000 + 148,000 + 52,200 = \$260,200$$

- **Facilities:** Facilities cost includes floor space, UPS, cooling infrastructure, fire suppression, and data center services. Scaled from LRZ's SuperMUC estimates, the total for 148 nodes is approximately €1.1M euros over 3 years [45]. Using a conversion rate of 1 EUR = 1.10 USD:

$$C_{\text{facilities}} = 1.1 \times 1,100,000 = \$1,210,000$$

- **Electricity:** SuperMUC-NG operates at approximately 427 watts for one node per hour at 90% load [11]. The estimated power consumption (with PUE = 1.06) for 148 nodes results in 67.0 kW. At €0.22/kWh [5], converted to USD, for 3 years:

$$\text{Annual Cost} = 67.0 \text{ kW} \times 8760 \text{ h} \times 0.242 = \$142,054 \Rightarrow \text{3-year total} = \$426,162$$

- **Hardware Replacement:** Assumed 5% annual failure/refresh rate for hardware capital [20]:

$$C_{\text{replacement}} = 0.05 \times \$1,936,000 \times 3 = \$290,400$$

- **Staffing:** 2 full-time equivalents (FTEs) for sysadmin at \$160,000/year combined:

$$C_{\text{staff}} = 160,000 \times 3 = \$480,000$$

- **Software Licensing:** Includes OS, storage software, and compilers [36]:

- SUSE Linux: $148 \times \$387 \times 3 = \$171,396$

- IBM GPFS: $148 \times \$500 \times 3 = \$222,000$

- Intel Compilers: $148 \times \$300 \times 3 = \$133,200$

$$C_{\text{software}} = 171,396 + 222,000 + 133,200 = \$641,200$$

Cost calculation from the formula in [20]:

$$C_{\text{total}} = C_{\text{fixed}} + C_{\text{variable}}$$

$$C_{\text{fixed}} = N \times C_{\text{node}} + C_{\text{storage}} + C_{\text{network}} + C_{\text{facilities}}$$

$$= 148 \times 12000 + 41600 + 260200 + 1210000 = \$3,287,800$$

$$C_{\text{variable}} = C_{\text{electricity}} + C_{\text{replacement}} + C_{\text{staff}} + C_{\text{software}}$$

$$= 426162 + 290400 + 480000 + 641200 = \$1,837,762$$

$$C_{\text{total}} = 3,287,800 + 1,837,762 = \$5,125,562$$

Total productive node-hours (3 years @ 90% utilization):

$$148 \times 26,280 \times 0.9 = 3,502,496 \text{ hours}$$

Per-node cost per hour:

$$\frac{\$5,125,562}{3,502,496} \approx \$1.46/\text{hour}$$